H Details of General Model Merging Baselines

In this section, we provide the detailed descriptions and formulations of the general-purpose model merging methods used in our comparison.

White-box Methods (Parameter-Access Required).

- **Task Arithmetic [11].** A simple yet effective method that linearly combines task vectors: $\theta_m = \theta_{\text{pre}} + \lambda \sum_t \tau_t$, where λ is a scaling factor controlling the strength of task integration.
- **TIES-Merging [27].** Addresses interference among tasks through a three-step procedure: **T**rimming low-magnitude parameters, **E**lecting a dominant sign across tasks, and **S**eparately merging parameters that agree in sign: $\tau_m = \text{TIES}(\{\tau_t\})$.
- **DARE (Drop and Rescale) [31].** Sparsifies task vectors by randomly **D**ropping a fraction p of delta parameters and **RE**scaling the remaining ones to preserve the expected magnitude: $\delta' = \frac{\delta \odot (1-m)}{1-p}$, where $m \sim \text{Bernoulli}(p)$.
- **DELLA-Merging [5].** Introduces MAGPRUNE, a magnitude-aware pruning strategy that preferentially drops small-magnitude deltas, followed by sign election and fusion: $p(\text{drop}_i) \propto \frac{1}{|\delta_i|}$.
- **EMR-Merging [9].** Constructs a unified model θ_{unified}, then recovers task-specific behavior via lightweight **M**asks (M_t) and **R**escalers (λ_t): $\theta'_t = \lambda_t \cdot (M_t \odot \theta_{\text{unified}})$.
- **Breadcrumbs [4].** Generates layer-wise sparse "breadcrumb" masks m_t that filter out both negligible and outlier weights from task vectors, enabling cleaner multi-task composition: $\theta_m = \theta_{\text{pre}} + \alpha \sum_t (m_t \odot \tau_t)$.
- **TSV-M [3].** Uses task singular vectors to identify and merge informative directions in task-vector space, requiring access to adapter parameters.
- **AdaMerging [29].** Learns adaptive merging coefficients for different model components using gradient-based optimization on validation data, and is therefore a white-box/gradient-based comparator rather than a black-box API-only method.
- **Robust Merging [12].** Improves robustness of parameter merging by reducing the influence of conflicting or noisy update directions, while still assuming direct access to model deltas.

Black-box Compatible Method.

- **LoRAHub [8].** Achieves cross-task generalization by dynamically composing LoRA modules $\{m_i\}$ via gradient-free optimization to learn combination weights $\{w_i\}$: $m_{\text{composed}} = \sum_i w_i m_i$.

I Hyperparameter Grid Search for Baselines

This section provides a summary of the hyperparameter grid search conducted for the baseline methods across all datasets. This table 9 presents the optimal configurations and their corresponding best performance (Precision and F1-score in %). The grid search for λ was conducted over the set $\{0.1, 0.3, 0.5, 0.7, 1.0\}$, and for the density p over $\{0.5, 0.7, 0.9, 1.0\}$.

J Qwen detail

To further validate our approach, we adopted the powerful Qwen model and evaluated on the InstructUIE benchmark [25], where Qwen outperforms alternatives like Llama3. This stronger base model improved all methods, including baselines. Nevertheless, `Evo-Merging` remains consistently superior. As shown in Table 12, it achieves the highest average Precision and F1 on in-domain tasks. Similarly, on out-of-domain (OOD) tasks (Table 13), it again attains the top average F1-score, confirming its effectiveness even with a more advanced base model.

J.1 Ablation Full Results

Table 9: Summary of hyperparameter grid search for baseline methods.

Dataset	TIES-Merging			DARE			Della			Breadcrumbs		
	Params (λ, p)	Prec.	F1	Params (λ, p)	Prec.	F1	Params (λ, p, ϵ)	Prec.	F1	Params (λ, p, γ)	Prec.	F1
NER-mitrestaurant	(1.0, 0.9)	21.40	24.22	(1.0, 0.9)	10.88	14.31	(0.5, 0.9, 0.1)	29.69	26.06	(0.5, 0.7, 0.01)	38.99	34.06
NER-FindVehicle	(1.0, 0.9)	29.87	29.45	(0.3, 0.9)	24.13	23.69	(0.7, 0.9, 0.1)	16.66	17.04	(0.5, 0.9, 0.01)	47.41	38.42
ET-mitmovie	(0.9, 0.9)	48.67	51.12	(0.1, 0.5)	44.57	48.60	(0.7, 0.9, 0.1)	43.85	47.68	(0.5, 0.7, 0.01)	49.73	51.60
ET-FabNER	(0.5, 0.9)	15.91	11.42	(0.1, 0.9)	15.44	10.71	(1.0, 0.9, 0.1)	14.41	12.51	(0.5, 0.9, 0.01)	20.53	16.84
RE-NYT	(1.0, 0.9)	12.55	7.57	(0.1, 0.9)	3.69	2.53	(0.5, 0.9, 0.1)	3.94	3.25	(0.5, 0.9, 0.01)	34.05	28.03
RE-CoNLL04	(1.0, 0.9)	20.78	19.05	(0.1, 0.9)	10.85	8.08	(1.0, 0.9, 0.1)	19.91	17.97	(0.5, 0.9, 0.01)	26.30	24.56

Table 10: Standardized Model Distribution Across Datasets. This table shows the source adapters selected by oracle top-K pre-selection for each target OOD task.

Target: MIT-Restaurant	Target: FindVehicle	Target: New York Times	Target: CoNLL04	Target: MIT-Movie	Target: FabNER
NER CrossNER AI	NER MIT-Movie SFT	NER CoNLL 2003	EPR CoNLL04	NER MIT-Movie	ET MultiNERD
NER CrossNER Literature	NER CoNLL 2003	EPR New York Times	EPR SciERC	NER MIT-Restaurant	ET CoNLL 2003
NER CrossNER Music	NER OntoNotes	RE NYT11	EPR New York Times	NER FindVehicle	ET MIT-Movie
NER CrossNER Politics	NER CrossNER Politics		RE SciERC	NER FabNER	ET CrossNER Science
NER CrossNER Science	NER Broad Tweet Corpus			NER CrossNER Literature	ET Broad Tweet Corpus
NER FabNER	NER WikiANN EN			ET CrossNER AI	ET OntoNotes
NER FindVehicle	NER CrossNER Science			NER CrossNER AI	ET WikiANN EN
NER MIT-Movie	NER TweetNER7			ET CrossNER Music	ET WikiNeural
NER TweetNER7	NER ACE 2005			NER CrossNER Music	ET FindVehicle
NER WikiANN EN	NER MultiNERD			ET CrossNER Literature	ET MIT-Restaurant

Table 11: Main Results for the Evaluation of In-Domain (ID) Robustness.

Method	ACE_2005		CrossNER_AI		CrossNER_Lit		FabNER		Mit-Movie		MultiNERD		TweetNER		WikiANN		Avg	
	Prec	F1	Prec	F1	Prec	F1	Prec	F1	Prec	F1	Prec	F1	Prec	F1	Prec	F1	Prec	F1
TIES	8.33	6.26	4.56	3.82	4.69	3.82	7.00	5.55	6.02	5.14	20.93	22.87	10.73	9.94	40.27	38.99	12.82	12.05
DARE	5.62	4.48	1.89	1.51	1.41	1.28	3.54	2.93	2.62	2.17	11.63	12.55	5.61	5.23	31.44	30.52	7.97	7.58
DELLA	5.65	4.47	1.52	1.12	1.46	1.25	3.88	3.11	2.52	2.17	12.13	13.18	5.83	5.32	31.11	30.13	8.01	7.59
Breadcrumbs	17.04	18.15	21.56	14.90	30.82	27.09	11.00	6.06	20.80	15.73	34.03	47.72	27.07	22.90	43.06	39.36	25.67	23.99
Task Arithmetic	16.42	17.67	22.61	14.15	30.89	25.48	11.03	5.47	20.29	14.47	36.10	49.15	26.88	22.28	44.35	40.14	26.07	23.60
EMR-Merging	16.34	17.90	20.97	13.02	29.12	24.81	11.70	5.80	20.47	14.53	35.65	49.08	26.26	21.25	43.28	39.11	25.47	23.19
LoRAHub	23.85	20.59	32.76	32.44	35.56	32.72	26.45	22.48	35.62	33.12	35.46	42.15	25.12	27.27	52.70	54.12	33.44	33.11
Evo-Merging+	**28.62**	**22.58**	**42.03**	**41.03**	**40.31**	**37.82**	**35.71**	**31.27**	**56.77**	**52.34**	**44.68**	**51.00**	**37.41**	**38.16**	**63.20**	**63.40**	**43.59**	**42.20**

Table 12: Main results of Qwen2.5 7B with selected baselines over in-domain tasks. † indicates that the improvement of `Evo-Merging` over the strongest baseline (LoRAHub) is statistically significant ($p < 0.05$).

Method	ACE_2005		CrossNER_AI		CrossNER_Lit		FabNER		Mit-Movie		MultiNERD		TweetNER		WikiANN		Avg	
	Prec	F1	Prec	F1	Prec	F1	Prec	F1	Prec	F1	Prec	F1	Prec	F1	Prec	F1	Prec	F1
Individual model	74.74	74.05	37.31	34.11	44.54	41.96	52.86	50.25	83.65	82.86	89.06	88.70	60.68	58.00	84.89	84.67	65.97	64.32
TIES	27.32	21.71	43.25	39.67	**43.85**	**40.41**	25.20	19.79	58.27	56.62	53.00	60.14	35.61	36.11	56.38	56.28	42.86	41.34
DARE	26.98	21.53	43.82	40.23	43.74	40.26	26.10	20.54	58.52	56.89	53.29	60.41	35.27	35.98	56.70	56.69	43.05	41.57
DELLA	26.93	21.43	43.77	40.22	43.55	39.99	26.37	20.75	58.98	**57.32**	53.20	60.25	35.31	36.05	56.67	56.58	43.10	41.57
Breadcrumbs	26.40	21.11	42.09	38.87	42.58	39.58	23.98	18.92	57.21	55.78	51.75	58.61	34.78	35.30	54.89	54.85	41.71	40.38
Task Arithmetic	26.83	21.29	43.17	39.63	43.54	40.11	25.48	20.01	57.84	56.27	53.15	60.24	34.69	35.12	55.75	55.79	42.56	41.06
LoRAHub	**31.17**	**23.24**	35.79	28.77	35.81	30.32	17.31	12.11	54.94	49.17	60.46	64.58	41.82	36.92	64.09	62.26	42.67	38.42
EMR-Merging	27.19	21.78	43.66	40.38	42.76	39.52	25.55	20.05	57.77	56.33	52.71	59.91	34.71	35.52	55.66	55.71	42.50	41.15
`Evo-Merging`	25.90	19.62	**49.02**	**44.08**	39.14	34.84	**28.08**	**21.34**	**62.67**	55.46	**62.55**	**67.41**	**43.52**	**41.25**	**64.59**	**63.10**	**46.93**	**43.39**†

Table 13: Main results of our methods and the selected baselines over out-of-domain (OOD) tasks.

Method	NER				RE				ET				Avg	
	Mit-Restaurant		FindVehicle		New York Time		Conll04		Mit-Movie		FabNER			
	Prec	F1	Prec	F1	Prec	F1	Prec	F1	Prec	F1	Prec	F1	Prec	F1
TIES	61.78	48.34	23.81	20.98	32.64	29.79	19.70	17.77	58.47	47.13	50.91	33.17	41.22	32.86
DARE (TIES+)	58.97	45.75	16.65	14.29	31.95	29.28	18.83	16.96	50.51	39.88	48.98	31.72	37.65	29.65
Task Arithmetic	63.52	50.65	29.49	25.99	25.28	23.24	19.70	17.82	59.60	47.08	50.50	32.21	41.35	32.83
Breadcrumbs	63.15	50.89	42.94	39.68	24.21	22.01	22.29	19.91	62.61	51.18	49.58	32.43	44.13	36.02
Della	59.05	45.82	16.93	14.58	31.30	28.45	18.40	16.52	51.56	40.81	49.05	31.78	37.72	29.66
EMR-Merging	60.82	48.18	24.78	21.32	43.16	38.60	19.70	17.77	58.23	45.51	50.41	31.15	42.85	33.76
Best Adapter	63.66	57.64	59.48	56.67	63.23	55.52	33.12	31.21	64.15	63.72	**53.07**	**50.28**	56.12	52.51
LoRA	59.27	56.83	52.87	51.62	24.32	22.75	33.12	31.21	59.27	56.83	11.13	10.70	40.00	38.32
LoRAHub	56.20	52.13	**71.64**	**70.86**	**61.17**	**61.15**	31.23	29.77	**77.96**	**77.06**	45.45	46.72	57.28	56.28
Evo-Merging	**70.86**	**69.55**	69.70	69.36	60.24	58.95	**47.29**	**44.19**	75.20	75.36	51.39	49.97	**62.45**	**61.23**

Table 14: Full per-task ablation study of Evo-Merging's components. [↓] means the performance drop relative to our full model.

Method	NER_Mit-Rest		NER_FindVeh		RE_NYT		RE_Conll04		ET_Mit-Movie		ET_FabNER		Avg	
	Prec	F1	Prec	F1	Prec	F1	Prec	F1	Prec	F1	Prec	F1	Prec	F1
Evo-Merging	**66.29**	**65.99**	**49.61**	**47.64**	**57.81**	**51.15**	**41.87**	**43.00**	**64.11**	**62.55**	**43.10**	**42.47**	**53.80**	**52.13**
w/o Denoising & Scaling	28.01	33.10	44.14	32.68	2.35	1.94	7.36	5.37	43.70	39.67	14.10	11.75	23.28 [↓ 30.5]	20.75 [↓ 31.4]
w/o Denoising	65.05	63.08	31.81	30.75	54.23	48.61	21.30	21.77	41.99	41.70	41.94	39.74	42.72 [↓ 11.1]	40.94 [↓ 11.2]
w/o Scaling	39.90	37.41	48.78	41.35	2.27	2.23	7.14	6.38	46.10	46.52	14.10	11.75	26.38 [↓ 27.4]	24.27 [↓ 27.9]
w/o Sign-flipped	42.04	37.70	46.61	41.07	45.45	38.38	35.21	32.91	62.19	60.47	34.77	29.80	44.38 [↓ 9.4]	40.06 [↓ 12.1]